

A DEGREE-FOUR STAR-BODY COUNTEREXAMPLE TO A VOLUMETRIC QUESTION FOR 2-INTERSECTION BODIES

AGENT LANE 17

STATUS

This packet gives a candidate counterexample to Question 3 in Koldobsky–Paouris–Zymonopoulou, *Isomorphic properties of intersection bodies* [1]. The question asks whether every symmetric star body K of volume 1, whose k -intersection body $I_k(K)$ exists, satisfies

$$|I_k(K)| \leq |I_k(D_n)|,$$

where D_n is the Euclidean ball of volume 1. The answer is negative already for $n = 3$ and $k = 2$.

SOURCE QUESTION

The source paper defines D_n to be the Euclidean ball of volume 1 and asks:

Is it true that if K is a symmetric star body of volume 1, $1 \leq k \leq n - 1$, and if $I_k(K)$ exists, then

$$|I_k(K)| \leq |I_k(D_n)| \text{ ?}$$

Note that a positive answer to the previous question would easily imply a linear in k estimate in Theorem 1.2.

The second conclusion of Theorem 1.3. can be viewed as a generalization of the classical Busemann inequality (see Introduction). We don't know if the assumption that $I_k(K)$ is convex is necessary in Theorem 1.3 and whether the estimate can be replaced by 1:

Question 3: Is it true that if K is a symmetric star body of volume 1, $1 \leq k \leq n - 1$ and if $I_k(K)$ exists, then

$$|I_k(K)| \leq |I_k(D_n)| \text{ ?}$$

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DEFINITIONS AND CONVENTIONS

For star bodies $K, D \subset \mathbb{R}^n$, the source paper says that D is the k -intersection body of K if, for every $(n - k)$ -dimensional subspace $H \subset \mathbb{R}^n$,

$$|D \cap H^\perp| = |K \cap H|.$$

When such a body exists it is denoted $I_k(K)$.

In this packet $n = 3$ and $k = 2$. Thus H is a line and $H^\perp$ is the perpendicular plane. Let σ denote normalized surface measure on S^2 . For $\xi \in S^2$, write

$$(Rf)(\xi) = \int_{S^2 \cap \xi^\perp} f(u) \, du$$

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for the unnormalized Funk transform, where du is arclength measure on the great circle.

PROOF INTUITION

For $n = 3, k = 2$, the 2-intersection condition is a Funk-transform inversion problem: the plane section areas of $I_2(K)$ are obtained by applying R to the square of the radial function of $I_2(K)$. The degree-four zonal spherical harmonic P_4 is an eigenfunction of R with positive eigenvalue $2\pi P_4(0) = 3\pi/4$. This lets us choose K with radial function $1 + \varepsilon P_4$ and choose $I_2(K)$ explicitly with squared radial function $1 + (8/3)\varepsilon P_4$.

The volume normalization slightly shrinks K , but the volume of the constructed $I_2(K)$ changes by the factor

$$(\mathbb{E}(1 + \varepsilon P_4)^3)^{-1/2} \mathbb{E} \left(1 + \frac{8}{3} \varepsilon P_4 \right)^{3/2}.$$

The first nonzero term in its Taylor expansion is $7\varepsilon^2/54 > 0$. Hence the Euclidean ball is not a maximizer among star bodies whose 2-intersection body exists.

COUNTEREXAMPLE

Theorem 1. *Question 3 of [1] is false as stated. More precisely, for all sufficiently small $\varepsilon > 0$, there is an origin-symmetric star body $K_\varepsilon \subset \mathbb{R}^3$ with $|K_\varepsilon| = 1$ such that $I_2(K_\varepsilon)$ exists and*

$$|I_2(K_\varepsilon)| > |I_2(D_3)|.$$

Proof. Let

$$P(t) = P_4(t) = \frac{35t^4 - 30t^2 + 3}{8}.$$

This is the degree-four Legendre polynomial. For small $\varepsilon > 0$, define

$$A(\varepsilon) = \int_{S^2} (1 + \varepsilon P(u_3))^3 d\sigma(u)$$

and let $r = (3/(4\pi))^{1/3}$, so $D_3 = rB_2^3$. Put

$$a_\varepsilon = rA(\varepsilon)^{-1/3}, \quad \rho_{K_\varepsilon}(u) = a_\varepsilon(1 + \varepsilon P(u_3)).$$

Since P is even and bounded below, this is the radial function of an origin-symmetric star body for all sufficiently small ε . Its volume is

$$|K_\varepsilon| = \frac{4\pi}{3} \int_{S^2} \rho_{K_\varepsilon}(u)^3 d\sigma(u) = \frac{4\pi}{3} a_\varepsilon^3 A(\varepsilon) = \frac{4\pi}{3} r^3 = 1.$$

Now define a second star body D_ε by

$$\rho_{D_\varepsilon}(u)^2 = \frac{2a_\varepsilon}{\pi} \left(1 + \frac{8}{3} \varepsilon P(u_3) \right).$$

This is positive for all sufficiently small $\varepsilon > 0$. We claim $D_\varepsilon = I_2(K_\varepsilon)$. The Funk-Hecke formula gives, for the degree-four zonal harmonic,

$$R(P(u_3))(\xi) = 2\pi P(0)P(\xi_3) = \frac{3\pi}{4}P(\xi_3),$$

and $R\mathbf{1} = 2\pi$. Hence

$$R(\rho_{D_\varepsilon}^2)(\xi) = \frac{2a_\varepsilon}{\pi} \left(2\pi + \frac{8}{3} \varepsilon \frac{3\pi}{4} P(\xi_3) \right) = 4a_\varepsilon(1 + \varepsilon P(\xi_3)).$$

If $H = \text{span}\{\xi\}$, the area of the central plane section of D_ε perpendicular to H is

$$|D_\varepsilon \cap H^\perp| = \frac{1}{2} R(\rho_{D_\varepsilon}^2)(\xi) = 2a_\varepsilon(1 + \varepsilon P(\xi_3)).$$

The section of K_ε by the line H has length

$$|K_\varepsilon \cap H| = 2\rho_{K_\varepsilon}(\xi) = 2a_\varepsilon(1 + \varepsilon P(\xi_3)),$$

because K_ε is origin-symmetric. Therefore $|D_\varepsilon \cap H^\perp| = |K_\varepsilon \cap H|$ for every line H , so the 2-intersection body $I_2(K_\varepsilon)$ exists and equals D_ε .

It remains to compare volumes. For the Euclidean ball $D_3 = rB_2^3$, its 2-intersection body is the Euclidean ball of radius

$$s = \sqrt{\frac{2r}{\pi}},$$

since a central plane section of the ball sB_2^3 has area $\pi s^2 = 2r$, the length of a central line section of rB_2^3 . Thus

$$|I_2(D_3)| = \frac{4\pi}{3} \left(\frac{2r}{\pi} \right)^{3/2}.$$

On the other hand, with

$$B(\varepsilon) = \int_{S^2} \left(1 + \frac{8}{3} \varepsilon P(u_3) \right)^{3/2} d\sigma(u),$$

we have

$$|D_\varepsilon| = \frac{4\pi}{3} \left(\frac{2a_\varepsilon}{\pi} \right)^{3/2} B(\varepsilon).$$

Therefore

$$\frac{|I_2(K_\varepsilon)|}{|I_2(D_3)|} = A(\varepsilon)^{-1/2} B(\varepsilon).$$

The normalized spherical moments of $P_4(u_3)$ are

$$\mathbb{E}P = 0, \quad \mathbb{E}P^2 = \frac{1}{9}, \quad \mathbb{E}P^3 = \frac{18}{1001}.$$

Consequently

$$A(\varepsilon) = 1 + \frac{\varepsilon^2}{3} + \frac{18}{1001} \varepsilon^3.$$

Taylor expansion also gives

$$B(\varepsilon) = 1 + \frac{8}{27} \varepsilon^2 + O(\varepsilon^3).$$

Hence

$$A(\varepsilon)^{-1/2} B(\varepsilon) = 1 - \frac{\varepsilon^2}{6} + \frac{8}{27} \varepsilon^2 + O(\varepsilon^3) = 1 + \frac{7}{54} \varepsilon^2 + O(\varepsilon^3).$$

The coefficient $7/54$ is positive, so for every sufficiently small positive ε ,

$$\frac{|I_2(K_\varepsilon)|}{|I_2(D_3)|} > 1.$$

This contradicts the proposed inequality in Question 3. □

VERIFICATION NOTES

- Local run indexes were searched for 1105.2629, the paper title, `I_k(K)`, `I_k(D_n)`, **Question 3**, and intersection-body keywords. No duplicate packet for this target was found.
- Bounded web search on 2026-07-03 used the exact title, **Question 3**, `I_k(K)`, `I_k(D_n)`, **Busemann inequality**, **k-intersection body**, and **star body**. It found the source arXiv page [1] and no separate later paper explicitly resolving Question 3.
- The script `code/verify_moments.py` checks the exact rational moments of P_4 and the Taylor coefficients used above. This code is only an arithmetic audit; the proof itself is the analytic Funk-transform construction above.

LIMITATIONS

This is a counterexample to the star-body formulation in Question 3. It does not contradict the source paper's Theorem 1.3 with the additional convexity assumption on $I_k(K)$, and it does not decide whether a Euclidean maximization statement survives under extra convexity hypotheses on K or on $I_k(K)$.

HUMAN REVIEW RECOMMENDATION

Review as a likely-valid counterexample to Question 3. The main checks are: the normalization of the 2-intersection-body definition for $n = 3, k = 2$, the Funk-transform eigenvalue $R(P_4) = 2\pi P_4(0)P_4$, and the second-order volume expansion $1 + 7\varepsilon^2/54 + O(\varepsilon^3)$.

REFERENCES

- [1] A. Koldobsky, G. Paouris, and M. Zymonopoulou, *Isomorphic properties of intersection bodies*, arXiv:1105.2629, 2011. <https://arxiv.org/abs/1105.2629>